

STAT 112

INTRODUCTION TO STATISTICS AND PROBABILITY II

SESSION 3A- Introduction to Probability Distributions 1

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Session Outline

- 1 Probability Distributions
- 2 Discrete Probability Distribution
- 3 Continuous Probability Distribution

Session Overview

In the previous sessions, the basic concepts and relative frequency definition of probability were presented and discussed. In this session, definition of probability distribution of random variables will be discussed. Students will be exposed to conditions a function must satisfied to be known as probability distribution and probability distributions for discrete and continuous random variables.

Learning Outcomes

- At the end of the session, students will be able to:
 - 1 Define and explain probability distribution of a random variable.
 - 2 Demonstrate that a function of a random variable is a probability distribution.
 - 3 Distinguished between probability mass function and probability density function.
 - 4 Construct a probability distribution for a discrete random variable.

Definition of Probability Distribution

- A **Probability distribution** is a statistical function that describes all the possible values and likelihoods that a **random variable** can take within a given range.
- Alternatively: A probability distribution is a representation of random variables and the associated probabilities of different outcomes.
- A probability distribution is characterized by a probability density function or probability mass function for continuous or discrete variables, respectively.

Example of Probability Distribution

- In an experiment of tossing a fair coin twice. The sample space $\Omega = \{TT, TH, HT, HH\}$. If we define the random variable X to be the number of heads observed in the two tosses, then we have X assuming the values 0, 1, 2.
- The associated probabilities of the possible values of the random variable can be summarize in a tabular form as

X	0	1	2
$P(X = x)$	0.25	0.5	0.25

Discrete Probability Distribution

- A distribution is called a discrete probability distribution, where the set of outcomes are discrete in nature.
- A tabular, formula or graph that lists all possible values a discrete random variable can assume, together with their associated probabilities, is called a **discrete probability distribution**.
- For example, if a dice is rolled, then all the possible outcomes are discrete and give a mass of outcomes. This is also known as probability mass functions.

Discrete Probability Distribution

- 1 A typical discrete probability distribution in a table form is shown as follows:

X	x_1	x_2	x_3	$\cdots$	x_n
$P(X = x_i)$	$p(x_1)$	$p(x_2)$	$p(x_3)$	$\cdots$	$p(x_n)$

- 2 **Example:** The probability distribution of the number of cars a dealer sells daily.

Daily sales: X_i	0	1	2	3	4
$P(X = x_i)$	0.05	0.15	0.35	0.25	0.20

- 3 Note: $\sum_{i=1}^4 p(x_i) = 1$

Discrete Probability Distribution

- 2 A typical discrete probability distribution in a formula form is shown as follows:

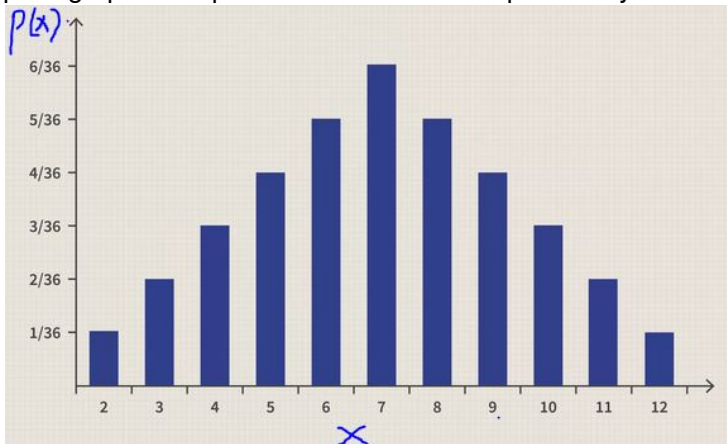
$$P(X = x_i) = \frac{1}{10}(x_i - 1), \quad \text{where } x_i = 1, 2, 3, 4, 5$$

Note that $\sum_{i=1}^5 P(X = x_i) = 1$.

- 2 Bar graph or histogram is another way of representing a discrete probability distribution. Consider the number observed when rolling two standard six-sided dice. Each die has a $1/6$ probability of rolling any single number, one through six, but the sum of two dice will form the probability distribution depicted in the graph below.

Discrete Probability Distribution

A typical graphical representation of discrete probability distribution:



Legitimacy Condition

If a discrete random variable X can take values x_i with associated probabilities $p(x_i)$, then the following conditions must be true or satisfied before it can be known as a probability distribution:

① $0 \leq p(x_i) \leq 1$, for all x_i .

② $\sum_{\text{all } x} p(x_i) = 1$,

where $p(x_i) = P(X = x_i)$ is known as *probability mass function*.

The collection of pairs $(x_i, p(x_i))_{i=1,2,3,\dots}$ is called the *discrete probability distribution function* of X .

Example 1

- Marketing estimates that a new instrument for the analysis of soil samples will be very successful, moderately successful, or unsuccessful, with probabilities 0.3, 0.6, and 0.1, respectively. The yearly revenue associated with a very successful, moderate successful, or unsuccessful product is GHs 10 million, GHs 5 million, and GHs 1 million, respectively. Let the random variable X denote the yearly revenue of the product. Determine the probability mass function of X .

The possible values of X are: 10m, 5m and 1m with their corresponding probabilities $p(x_i)$ 0.3, 0.6, 0.1, respectively. We can summarize the probability distribution function of X in a table below:

X in millions	10	5	1
$p(x_i)$	0.3	0.6	0.1

- Hence the probability mass function is given by:
- $P(X = 10\text{million}) = 0.3$
- $P(X = 5\text{million}) = 0.6$
- $P(X = 1\text{million}) = 0.1$
- It is clear that, $p(x_i) > 0$ for all x_i and $\sum_{i=1}^3 p(x_i) = 1$.

Example 2

- A discrete random variable Y has probability mass function given by

$$P(X = x) = \left(\frac{1}{2}\right)^x, \quad \text{where } x = 1, 2, 3, \dots$$

- Show that the probability mass function is legitimate and find the probability that X is even.

Solution

For the probability mass function to be legitimate, we check the following conditions

- ① $0 \leq p(x) \leq 1$: It is clear that $p(1) = \frac{1}{2}$ and $p(\infty) = 0$.

Therefore, $p(x) \geq 0$.

- ② Next, $\sum_{i=1}^{\infty} P(X = x) = \sum_{i=1}^{\infty} \left(\frac{1}{2}\right)^x = \frac{1/2}{1 - 1/2} = 1$

Hence the $P(X = x)$ is a legitimate probability mass function.

The $P(X = \text{even}) = P(X = k)$, where $k = 2, 4, 6, 8, 10, \dots$

Therefore,

$$P(X = \text{even}) = P(X = 2) + P(X = 4) + P(X = 6) + \dots$$

$$= \sum_{i=1}^{\infty} \left(\frac{1}{2}\right)^{2i} = \frac{1/4}{1 - 1/4}, \quad \text{where } i = 1, 2, 3, \dots$$

$$= \frac{1}{3}.$$

Continuous Probability Distribution

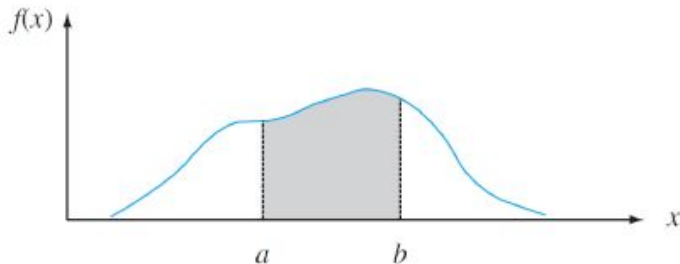
- **Definition:** Let X be a continuous random variable. Then a probability distribution or probability density function (*pdf*) of X is a function $f(x)$ such that for any two numbers a and b with $a \leq b$, we have

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$

- The probability that X is in the interval $[a, b]$ can be calculated by integrating the probability density function of the random variable X .
- In this case we cannot compute probabilities for individual values i.e. $P(X = a)$, but instead we assign probabilities to intervals of values i.e. $P(a < X < b)$.

Example: Graphical Representation

- The probability that X takes on a value in the interval $[a, b]$ is the area above this interval and under the graph of the density function $f(x)$:



$P(a \leq X \leq a)$ = the area under the density curve between a and b .

Legitimacy Condition

- The function $f(x)$ is called a probability density function (*pdf*) of the random variable X , if it satisfy the following conditions:
 - 1 $f(x) \geq 0$ for all values of x ,
 - 2 $\int_{-\infty}^{\infty} f(x)dx = 1$.

Example 3

Consider the continuous random variable X with probability density function

$$f(x) = \begin{cases} 2x, & 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$$

- 1 Show that the density function $f(x)$ is legitimate.
- 2 Calculate the following probabilities:
 $P(X \leq 0.4)$,
 $P(0.2 < X < 0.4)$ and
 $P(X < 0.5 | 1/3 < X < 2/3)$.

Solution

- ① Given that $f(x) = 2x$ for $0 < x < 1$,

It is clear that $f(x) \geq 0$ for all x , since $f(0) = 2(0) = 0$ and $f(1) = 2(1) > 0$. Thus, for any value of $x : 0 < x < 1$, $f(x) = 2x \geq 0$.
Next,

$$\int_0^1 f(x)dx = \int_0^1 2x dx = \left[\frac{2x^2}{2} \right]_0^1 = 1$$

Hence the density function is legitimate.

② a. $P(0.2 < X < 0.4) = \int_0^{0.4} 2x dx = x^2 \Big|_0^{0.4} = 0.16$

b. $P(0.2 < X < 0.4) = \int_{0.2}^{0.4} 2x dx = x^2 \Big|_{0.2}^{0.4} = 0.12$

c.

$$\begin{aligned} P(X < 0.5 | 1/3 < X < 2/3) &= \frac{P(X < 0.5, 1/3 < X < 2/3)}{P(1/3 < X < 2/3)} \\ &= \frac{P(1/3 < X < 1/2)}{P(1/3 < X < 2/3)} = \frac{5}{12} \end{aligned}$$

- ① Find the constant c such that the function

$$f(x) = \begin{cases} cx^2, & 0 < x < 3 \\ 0, & \text{elsewhere} \end{cases}$$

is a density function, and compute $P(1 < X < 2)$.

- ② Suppose that in a certain region the daily rainfall (in inches) is a continuous random variable X with probability density function $f(x)$ given by

$$f(x) = \begin{cases} \frac{3}{4}x(2-x), & 0 < x < 2 \\ 0, & \text{elsewhere} \end{cases}$$

Compute the probability that on a given day in this region the rainfall is

- ① not more than 1 inch.
- ② between $1/2$ and $3/2$ inches.

- 3 Verify that the following functions are probability mass functions, and determine the requested probabilities

i

X	-2	-1	0	1	2
$P(X = x)$	1/8	2/8	2/8	2/8	1/8

- a. $P(X \leq -2)$, b. $P(X > -2)$, c. $P(-1 < X < 1)$ and
d. $P(X \leq -1 \cup X = 2)$

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$$P(X = x) = \frac{8}{7} \left(\frac{1}{2} \right)^x, \quad x = 1, 2, 3$$

- a. $P(X \leq 1)$, b. $P(X > 1)$, c. $P(X \leq 1 \text{ or } X > 1)$

- 4 The probability distribution of a discrete random variable X is shown below

X	0	1	2	3
$P(X = x)$	c	c^2	$c^2 + c$	$3c^2 + 2c$

where c is a constant. Calculate

- a the constant c .
- b $P(0 \leq X \leq 2)$.
- c $P(X > 1)$.

- 5 A box contains 3 red balls and 2 green balls. Two balls are selected at random with replacement. Find the probability distribution of the random variable X that measures the number of red balls selected and show that it is legitimate probability mass function.

THANK YOU!!!